Test Specimen of the Quasi-Static Test on an RC Frame Substructure with a Parallel Double-Stage Yielding Buckling-Restrained Brace (PDYBRB)

1. Specimen configuration

The test specimen was a single-bay, single-story reinforced concrete (RC) frame substructure equipped with a parallel double-stage yielding buckling-restrained brace (PDYBRB) in an inverted-V (chevron) arrangement. It was extracted from the bottom transverse frame of a five-story RC teaching building located in a seismic intensity VIII region (design basic acceleration 0.30 g, Site Class II, characteristic period 0.40 s) and designed at a 1:2 geometric scale, with adjustments made to accommodate the loading conditions of the structural laboratory.

The substructure comprised two RC columns (C1, C2), one RC beam (B1) with a cast-in floor slab, a stiff RC foundation beam (B2), and the PDYBRB connected to the main frame through welded gusset plates and embedded steel parts. Longitudinal reinforcement areas of the frame beam and columns were obtained by scaling the results of a response-spectrum (mode-superposition) analysis of the prototype structure; transverse reinforcement and detailing followed GB 50010-2010.

2. Geometric dimensions

Table 1 Geometric dimensions of the frame substructure specimen

Component	**Cross-section b × h (mm)**	**Member length / other dimensions**	**Cover (mm)**
Frame column (C1, C2)	300 × 300	Total height 3150 mm; column-cap projection 50 mm	15
Frame beam (B1)	200 × 300	Beam-end projection 200 mm	15
Floor slab	thickness 70	Width 1040 mm; span 4300 mm	15
Foundation beam (B2)	550 × 600	Length 5750 mm	30

Notes: the clear span between column centerlines is 4300 mm and the total specimen width is 5750 mm. The slab overhang on each side of the beam web equals six times the slab thickness.

3. Reinforcement details

All reinforcing bars are hot-rolled ribbed bars of grade HRB400.

Table 2 Reinforcement details of the frame substructure specimen

Component	**Longitudinal bars**	**Ties**	**Confined-region length**	**Tie spacing (mm) confined / unconfined**
Frame column (C1, C2)	10 Φ18	Φ8	bottom 800 mm; top 500 mm	50 / 100
Frame beam (B1)	6 Φ16 (3 top + 3 bottom)	Φ8	500 mm at each end; 1000 mm at mid-span	50 / 100
Floor slab	Φ6 @ 100 mm, two-way	—	—	—
Foundation beam (B2)	14 Φ25	Φ8	—	100 (uniform)

Notes: the two columns are identically reinforced. Longitudinal bars of the beam and columns use 90° hooked anchorage, since the foundation-beam depth does not satisfy the straight-anchorage requirement.

4. Material properties

The specimen was cast with C40 ready-mixed concrete and cured for 28 days under the same conditions as the companion material specimens. Cube strengths were determined per GB/T 50081-2002 and reinforcement tensile properties per GB/T 228.1-2021. Measured properties are summarized in Table 3.

Table 3 Measured material properties

Material	**f_y (MPa)**	**f_u (MPa)**	**E ($\times 10^5$ MPa)**	**Elongation (%)**
Concrete C40 *	—	46.0	—	—
Steel Φ6	694.4	903.6	2.24	9.06
Steel Φ8	659.3	1045.2	2.20	9.28
Steel Φ16	637.8	845.5	2.20	12.39
Steel Φ18	524.3	713.9	2.02	13.50
Steel Φ25	474.1	643.9	1.98	13.94

* For concrete, the value listed is the mean cube compressive strength of three 150 mm cubes (45.7 / 43.6 / 48.8 MPa).